

# **Laplace for Second-Order Denominators and Intro to TFs**

**EGR 345**

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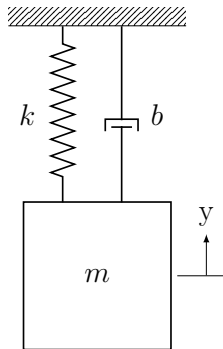
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# Outline

- quick review
- Laplace for second-order denominators
- Two things we need for lab:
  - transfer functions
  - plants

# Mass/Spring/Damper Response

- how does this system respond to an initial displacement?
  - by that I mean, find  $y(t)$
- what is the differential equation associated with this system?



## What is $\zeta$ ?

$$ms^2 + bs + k = (s^2 + 2\zeta\omega_n s + \omega_n^2)/m$$

$$\frac{b}{m} = ?$$

$$\frac{k}{m} = ?$$

## Finishing Up Second-Order Laplace Denominators

$$Y(s) = \frac{y_0(s + b/m)}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

## Laplace with complex 2<sup>nd</sup> order roots

**Key Idea:** if we have complex roots, we must use rows 8 **and** 9

$$Y(s) = \frac{y_0(s + b/m)}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1\omega}{(s+a)^2 + \omega^2} + \frac{C_2(s+a)}{(s+a)^2 + \omega^2}$$

# Transfer Functions and Plants

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# What is a transfer function?

- my software (which we will start using in lab) is designed around transfer functions and plants



# What is a plant?